Problem 1

A point charge $-Q$ (with $Q > 0$) is embedded at the center of a solid sphere of radius R. The sphere is made of a linear, isotropic, homogeneous dielectric with electric susceptibility χ_e. There is no other free charge anywhere.

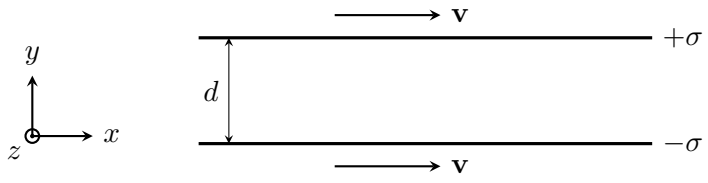
(a) Find the electric displacement $\mathbf{D}$ inside the dielectric ($0 < r \leq R$). Then find the electric field $\mathbf{E}$ in the same region.

(b) Find the polarization $\mathbf{P}$ inside the dielectric.

(c) Find the bound surface charge density σ_b on the outer surface of the sphere ($r = R$) and the bound volume charge density ρ_b everywhere inside the sphere. *Be careful at the origin.*

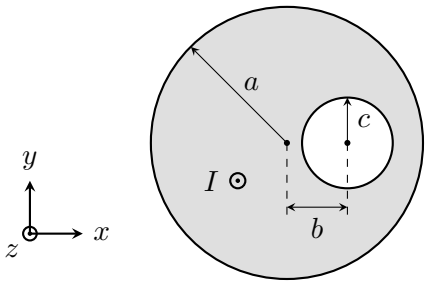
Problem 2

A large parallel-plate capacitor consists of two infinite, thin plates parallel to the xz-plane. The lower plate lies in the plane $y = 0$ and carries uniform surface charge density $-\sigma$. The upper plate lies in the plane $y = d$ and carries uniform surface charge density $+\sigma$ (with $\sigma > 0$). Both plates move together with constant velocity $\mathbf{v} = v\,\hat{\mathbf{x}}$, as shown below ($\hat{\mathbf{z}}$ points out of the page).



(a) Find the magnetic field $\mathbf{B}$ (magnitude and direction) in the region between the plates, $0 < y < d$.

(b) A long, straight cylindrical wire of radius a has a cylindrical hole of radius c drilled through its entire length. The axis of the hole is parallel to the axis of the wire and displaced from it by a distance b along $\hat{\mathbf{x}}$, with $b + c < a$. The wire carries a total steady current I in the $+\hat{\mathbf{z}}$ direction (out of the page), distributed uniformly over the remaining (shaded) cross-section.



Find the magnetic field $\mathbf{B}$ (magnitude and direction) at a point inside the hole.

Problem 3

An infinitely long straight wire of radius R lies along the z-axis. It carries a steady current with volume current density

$$\mathbf{J}(s) = k\, s^2\, \hat{\mathbf{z}}, \qquad 0 \le s \le R,$$

where s is the perpendicular distance from the axis and k is a positive constant.

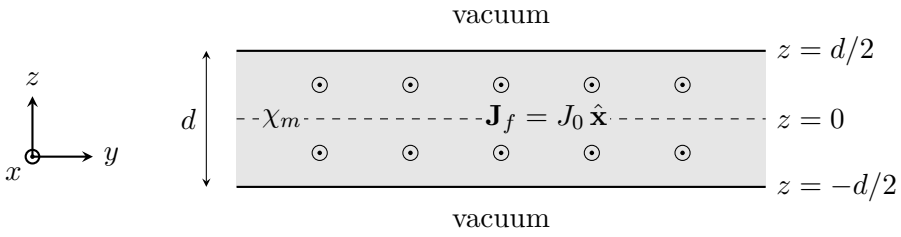
Find the magnetic vector potential $\mathbf{A}$ (magnitude and direction) inside the wire, $0 \le s \le R$. Take $\mathbf{A} = 0$ on the axis.

Problem 4

An infinite slab of linear magnetic material with magnetic susceptibility χ_m occupies the region $-d/2 \le z \le d/2$ and extends to infinity in the x and y directions. The slab carries a uniform free current density

$$\mathbf{J}_f = J_0\, \hat{\mathbf{x}},$$

where $J_0 > 0$ is a constant. The region outside the slab is vacuum. The figure shows a cross-section in the yz-plane, with $\hat{\mathbf{x}}$ (and the free current) pointing out of the page.



(a) Find the auxiliary field $\mathbf{H}$, the magnetization $\mathbf{M}$, and the magnetic field $\mathbf{B}$ inside the slab, $-d/2 < z < d/2$. Give both magnitude and direction.

(b) Find the bound volume current density $\mathbf{J}_b$ inside the slab and the bound surface current density $\mathbf{K}_b$ on the top ($z = d/2$) and bottom ($z = -d/2$) surfaces.